Prediction of mobile APP Advertising Conversion Rate Based on Machine Learning

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***Abstract*—With the rapid development of smart phones, mobile advertising has taken up more than half of the advertising market and has great potential in the future. At present, there are many mature researches on the conversion rate of search advertising, but few studies have been done on the conversion rate of mobile APP advertising. This paper studies the preliminary data of Tencent's first social advertising algorithm competition and provides a method for prediction of mobile APP advertising conversion rate based on machine learning.**

Keywords-Machine Learning; Advertising Conversion Rate; Prediction; Feature Engineering

I. INTRODUCTION

Internet advertising has always occupied a place in the advertising market with its convenient delivery method and cheaper price than TV commercials. Compared with traditional advertising, one of the biggest advantages of Internet advertising is that the effect is measurable, that is, the behavior after the user seen the advertisement can be recorded. If these data could be attributed under certain rules, which means that advertisers can clearly know the promotion effect of each channel and the number of users after the multichannel promotion. They can also analyses the effects of various delivery strategies based on data to further refine operations.

With the rapid popularity of smartphones and tablets, the emancipation of traffic restrictions by Wi-Fi, 3G and 4G, and the new interactive experience brought by Apple store ecosystem, the number and types of App developing at an alarming rate. As of March 2017, the Apple Store has about 2.2 million apps available and about 2.8 million apps are available in the Google Play Store. Figure 1 shows the number of applications for the major app stores in 2017.

The huge number of applications have brought the rapid development of mobile advertising. As a new type of advertising media, mobile app advertising has banner ads, video ads, the information flow and other forms, can make full use of the narrow space, and fully understands the needs of users and realizes the accurate placement of advertisements according to the real-time behavior and historical operations of user groups. It can also combine the powerful functions of mobile phones, such as location-based services, to provide users with more accurate advertising information.

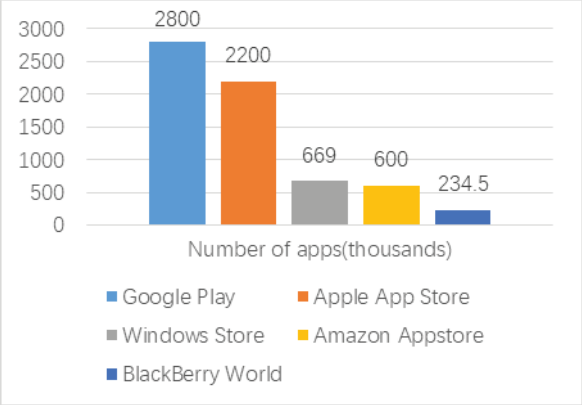

Figure 1. Number of apps in major app stores in 2017

According to the report of the Interactive Advertising Bureau (IAB) in the fourth quarter of 2016, the PC advertising revenue in 2016 was about $17.8 billion, accounting for 24% of the total advertising revenue. Mobile terminal revenue was $36.6 billion, accounting for 53% of the total revenue. Among them, mobile search is $17.2 billion, accounting for 47% of mobile advertising revenue, and mobile display related revenue of $18.1 billion, accounting for 49%, and other revenue of $1.3 billion, accounting for 4%. The advertising revenue in 2016 is shown in Figure 2. The proportion of advertising revenue in the past ten years is shown in Figure 3.

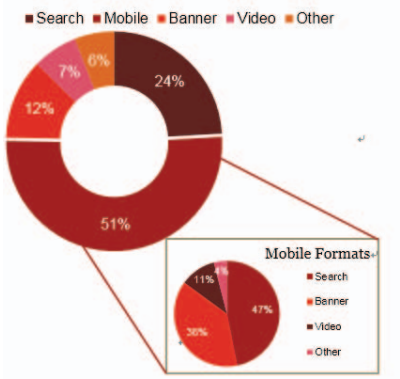

Figure 2. Advertising totalled revenue in 2016

According to the above data, we can see that mobile advertising occupies the largest share of the advertising market, and advantages of mobile advertising are doomed to continue. At present, there are many mature researches on

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the conversion rate of search advertising, but few researches have been done on mobile advertising.

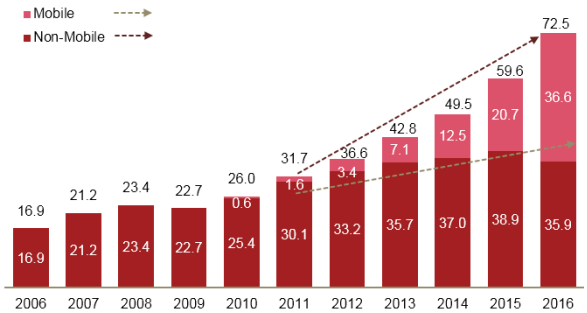

Figure 3. The proportion of advertising revenue in the last ten years

In April 2017, Tencent launched the first Tencent social advertising algorithm contest for college students, which requires competitors to predict the application download and activation rate after a user clicks on an ad based on a given data. This paper studies the conversion rate of mobile APP advertising by using the preliminary data obtained by Tencent.

II. PROBLEM DESCRIPTION

A. *Problem description*

Application activation is defined as the first time an application is opened after downloading. The competition calls for mobile app advertising as a research object to predict the probability of users activating app after clicking advertisements, that is, the probability of activation after clicking ads in a given advertisement, user and context. The data is provided after encrypting one month's log of Tencent's social platform, and the training set is the data from the 17th day to the 30th day of the month, the test set the data of the 31th day. The data distribution is shown in Figure 4.

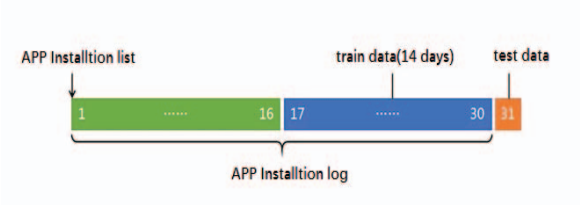

Figure 4. Diagram of Tencent data

B. *Data table interpretation*

Tencent's package includes eight data documents.

The ad table gives the features of advertisement, including advertisers, promotion plan, ad material, app ID, app platform. A promotion plan can contain multiple ads. Advertisers can put advertising with the same condition such as promotion platform and budget quota into the same promotion plan to manage. Multiple promotion plans, or advertisements can promote the same app at the same time. APP platform is valued as Android, IOS and unknown.

The User table shows user features, and contains user ID, user age, user gender, educational background, marital status, parenting status, user's hometown, permanent residence. User qualifications include primary school, junior high school, high school, specialist, undergraduate, master, doctor and unknown. Marital status includes single, newlywed, married and unknown. Parenting status refers to the status of the baby who is pregnant with a baby, which includes inoculation, has 0 to 6 months old baby, has 6 to 12 months old baby, 1 to 2 years old baby, 2-3 years old baby, and parenting but baby age unknown and unknown. The hometown refers to the birthplace of users, the value of which is specific to the city level city, using of second-level coding. And the thousands digit means the province, the tens digit and the single digits means the city.

The position table indicates the placement of the advertisement, includes the location of the ad, the site collection。

The user_app_actions table is users' app installation flow. APP installation flow refers to the recent user's behavior of app installation, including user ID, app ID and occurrence time.

The user_installedapps table is the list of app installation, which represents users' app installation list at a certain time point. Contains the user ID and app ID.

The app_categories table shows app classification features. The app classification uses 3-digit codes, and the hundreds digit represents the first class, the tens digit and the signal digit represent the secondary category, mark 0 means the class is unknown or cannot be obtained.

The training data was randomly sampled from a two consecutive weeks Tencent's social advertising system according to Apps and user's dimensions in the promotion from. Each training sample is an AD that clicks on the log.

The train file contains the tag, the time of the cl-ick, the return time, the material ID, user ID, ad's location ID, the networking mode, and the operator. The label is 0 indicating that there is no transformation after user clicks the ad, and 1 indicates that the transformation occurs after the click. When the la-bel is 0, the conversion time is an empty string. The Internet includes 2G, 3G, 4G, WIFI and unknown. Operators include China mobile, China Unicom, China telecom and unknown.

The test data was extracted from the AD log from the following day (i.e. 31 days) in the same sampling manner as the training data. Each line of test data file represents a test sample with instance ID, label, click time, material ID, user ID, location ID, networking mode, and operator. The instance ID uniquely identifies a sample, -1 represents the label placeholder usage, which represents the forecast.

C. *Difficulties in forecasting ad conversion rates*

As typical issues in advertising research, there are many similarities between click-through rate forecasting and conversion rate forecasting. They are both typical binary problems and there are many tools used in ctr forecasting can also be used in conversion forecasting. However, there are some difficulties only in forecasting ad conversion rate:

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(1) The characteristics of the conversion data are massive and sparse. In a large amount of data, the proportion of successful conversion is relatively small.

(2) The conversion data is noisy. Because the conversion data is reported by advertisers, there are many uncontrollable factors in the system and the reliability of data is relatively poor.

(3) The natural return time for conversion is long and the activation delay of mobile applications may take a few days. The reliability of data near the current time is relatively poor.

III. FEATURE ANALYSIS

In mobile advertising, attributes of app, user attributes, attributes of advertising and attributes of advertising platform together affect the conversion rate of applications. Based on the actual situation, we combined given data tables and get the following features.

A. *Basic properties*

Through preliminary statistical analysis of the training set, we can get a few very eye-catching features, networking methods and advertising positions. The conversion of each feature in the training set is shown in the Figure 5.

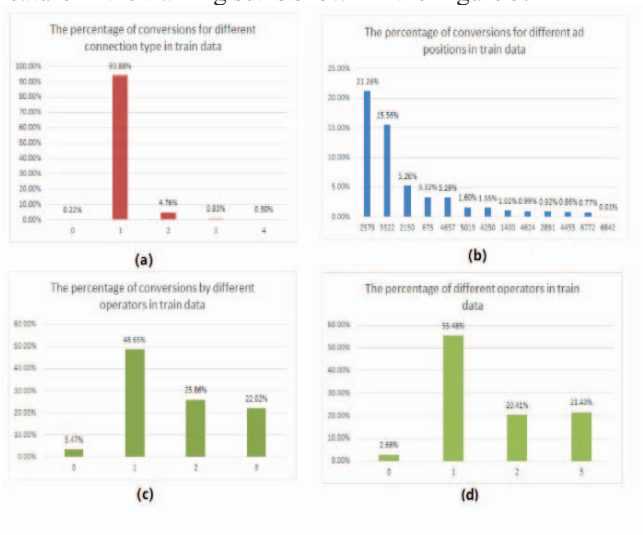

Figure 5. (a) The percentage of conversions for different connect types (b) The percentage of conversions for different ad position (c) The percentage of conversions by different operators (d) The percentage of different operators

1) *Networking features*

As shown in the Figure 5.3, the conversion amount is very large when the networking method is '1', accounting for 93.88% of the total historical conversion. According to the user's daily habits, it is not difficult to speculate that this method is Wi-Fi networking. The speed of Wi-Fi networking is generally faster than other networking methods and does not cost money. Therefore, the activation rate of app under the Wi-Fi network is much greater than other networking methods. The 4G network speed is fast, but it requires a certain cost, so the conversion ratio is only 4.76%, while the speed of 3G and 2G is poor, the possibility of users download apps is very small.

2) *Feature of advertising position*

The location of an advertising describes the location information of advertisements on different platforms. Whether an advertising position is visible or not, can be perceived by users at the first time, has a certain impact on the conversion rate of apps. Through the analysis of the data, we can find that the conversion rate of several advertising positions is far greater than other advertising positions, the maximum conversion ratio can reach 21.26%.

When selecting features, we should pay attention to real features, not only based on the conversion ratio. For example, the operator features, we can see from the Figure 5.3 that the conversion ratio of '1' is 48.65%, far greater than the other operators, but which mobile operator we used does not actually affect whether we decide to download one app in our life. We have analyzed all data, then we can clearly see from the Figure 5.4 that the '1' have accounted for 55.48% in all data, so the conversion ratio of '1' is higher probably caused by the higher numbers of users using '1' as operator, not a real feature.

B. *Statistical characteristics*

1) *Basic statistical characteristics*

The number of installed app and the count of similar installed app can be counted. For apps of the same category, if the user has installed more than one then proves that the user prefers this category, the conversion rate of other apps in the category is higher.

2) *Timing characteristics*

Some actions before a user clicks on an ad are of great significance for whether the click is converted. Using the number of users clicks before clicking time and the number of app installation can better analyze the behavior of users and predict the next behavior of users. For example, when a user is installing a large number of apps, the probability of clicking advertisement and then downloading the app is relatively high.

3) *Historical click features*

If the user clicks on the app advertisement in the near future without downloading it, and the user's network environment is not in the Wi-Fi environment, which means that the user is interested in that app, so the probability of downloading the app in Wi-Fi environment is relatively higher. But if the user has clicked on the app advertisement and in a Wi-Fi environment without downloading, and the amount of click is low, it may mean the user is not interested in the app, and the possibility of downloading this app in the future is low.

C. *Conversion features*

The requirement of this competition is to predict the conversion rate of mobile app ads after one user click, so the study of conversion rate is indispensable. Based on the research of the training data, we get the conversion features according to statistics: the conversion rate of ad, the conversion rate of advertisers, the conversion rate of advertising material, the conversion rate of app, the conversion rate of advertising platform, the conversion rate of advertising position, the conversion rate of user, the

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conversion rate of parenting status, the conversion rate of marital status, the conversion rate of gender, the conversion rate of education level and so on.

D. User similarity characteristics

Users may have the same interest in the same age, education level, or having the same marriage status and parenting state. More similar the users are, the higher probability of using the same app, so we can calculate the similarity between each user by processing the characteristics of users. Person correlation coefficient is used for similarity calculation, and the calculation formula of the person coefficient is as follows:

$$r(X,Y) = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2} \cdot \sqrt{n\sum y^2 - (\sum y)^2}} \quad (1)$$

In the calculation, users' age is adjusted according to the Chinese age division rule because the age value is too large than the other features, and then users is divided into 7 age groups:

$$\begin{cases} x < 12 & \text{age} = 0 \\ 12 < x < 18 & \text{age} = 1 \\ 18 < x < 28 & \text{age} = 2 \\ 28 < x < 40 & \text{age} = 3 \\ 40 < x < 55 & \text{age} = 4 \\ 55 < x < 65 & \text{age} = 5 \\ 65 < x < 80 & \text{age} = 6 \end{cases} \quad (2)$$

E. User-advertiser features

The number of app products of advertisers in the current installation list of users reflects the users' satisfaction with advertisers. If the number of products of the same advertiser in the list is not less than three, it indicates that the user is more satisfied with the products of the advertiser, and then the user has a higher probability of downloading this advertiser's app.

F. User-APP features

At the beginning of the design of the app, user group was set up. Different user groups have different interests. For example, the male group prefers game and sport apps, and the group with children tends to parenting apps. Since Tencent did not give a specific explanation on app classification, we combined apps with the user's gender, age, education level, marital status, parental status, and other characteristics to form user-app feature.

IV. MODEL TRAINING

A. Data processing

The behavior of the user launching the app requires to be reported back to the advertising system by advertiser, which usually has a certain delay. Advertisers report app activation data to the advertising system does not exceed 5 days. Data older than 5 days will be ignored by the system and not be counted in logs. As shown in Figure 3, the training data provided by Tencent in this competition is from the 17th day to the 30th day in a month. Therefore, for the last few days of train data, some unconverted data may be inaccurate, so we only use the data to the 26th day to ensure the accuracy of the data.

Because the label item is -1 in the test data given by Tencent and the correct test data has not been published, in order to evaluate our prediction, we used the training data of the 26th day as test data. And because the data is related to the time series, in order to accurately predict and prevent data leakage which means that future data is mixed into features, and then affect the prediction effect. We chose data from the 20th day to the 25th day as a train set.

For features with smaller dimensions, such as gender, education, connection type and remote operators, we use a one-hot code to process. For the missing value, it can be filled with the average value, for example, if the return time is missing, the historical average return time is used to fill in the missing value. We also perform Bayesian smoothing on the conversion characteristics.

1) Bayesian smoothing

In the actual data, using the conversion rate as a feature, you may encounter some problems. Due to the issue time of advertising, the statistics of new advertisings may be not sufficient, resulting in data errors, and because of the difference in the number of exposures between different advertisements, using conversion rate directly is not very reasonable. When using the user's historical conversion rate as a feature, since most users who have successfully converted only click on the advertisement once, the user's historical conversion rate is 100%, which will greatly affect the prediction of the conversion rate. At this time, we need to use Bayesian smoothing to process the data and use prior knowledge to give an initial value of the conversion rate.

The idea of Bayesian smoothing is to pre-set an empirical initial value for the conversion rate, and then correct this initial value by the current conversion amount and the number of clicks, as in [4]. If an ad's click number and conversion number are both zero, the conversion rate for that ad is the initial value. If the exposure amount between advertisement A and advertisement B is very different, it can be corrected by this initial value, so that the weight of the advertisement with a large exposure amount increases. Bayesian smoothing is the process of determining the empirical value.

B. XGBoost

XGBoost is a variant of GBDT. As in [3], the biggest difference is that xgboost does the second-order Taylor expansion of the objective function to calculate the leaf node weight of the tree to be fitted next (need to determine the structure of the tree first), then calculated the loss of each split node based on the loss function and finally the appropriate attribute to split according to the split loss.

This loss function derived from the second-order expansion is closely related to the process of splitting nodes. First, all the attributes of all nodes are traversed and split. Assuming that a certain attribute is selected as a split node, the weight of each leaf node of the tree can be calculated according to the formula developed by Taylor, and then the degree of loss reduction can be calculated. In this way, the feature that makes the loss reduction the most is selected as

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the splitting attribute of the current node. And so on until the termination condition is satisfied. In addition to the reduction factor similar to GBDT, xgboost penalizes the number of leaves and the weight of each tree to avoid overfitting.

In the recent data competition of big companies, the frequency of XGBoost has been increasing, which has brought good results to the participants.

C. LightGBM

LightGBM is a gradient boosting framework using decision tree based on learning algorithm. It can be said to be distributed, efficient, and has the following advantages: faster training efficiency, low memory use, higher accuracy, support for parallel learning and processing of large-scale data. The most important feature of LightGBM is the use of histogram algorithm and leaf growth strategy with depth constrained Leaf-wise.

The basic idea of histogram algorithm is to disperse successive floating-point eigenvalues into k integers and construct a histogram with a width of k. When the data is traversed, the statistics are accumulated in the histogram based on the discretized value as an index. After traversing the data, the histogram accumulates the required statistics, and then traverses to find the optimal segmentation point based on the discrete values of the histogram.

Level-wise data can split the leaves of the same layer at the same time. It is easy to perform multi-thread optimization, control model complexity， and not easily overfit. But in fact, Level-wise is an inefficient algorithm because it treats the leaves of the same layer indiscriminately, which brings a lot of unnecessary overhead. In fact, many leaves have a low split gain and there is no need to search and split.

Leaf-wise is a more efficient strategy. Each time leaves with the largest splitting gain are found from all leaves, then split. Therefore, compared with Level-wise, Leaf-wise can reduce more errors and achieve better accuracy under the same split times. The drawback of Leaf-wise is that a deeper decision tree may be produced, resulting in over fitting. Therefore, LightGBM adds a maximum depth limit on Leaf-wise, which ensures high efficiency while avoiding over fitting.

Currently, lightGBM has been favored by more and more people due to its high efficiency, low memory usage and high precision.

D. Predictive evaluation

Predictive assessments are performed using the Logarithmic Loss assessment, which has the following formula：

$$\log loss = -\frac{1}{N}\sum_{i=1}^{N}(y_i \log(p_i) + (1-y_i)\log(1-p_i)) \quad (3)$$

Where, N is the total number of test samples, and y_i is a binary variable. A value of 0 or 1 indicates the label of the i-th sample, and p_i is the probability that the model predicts that the i-th sample label is 1. The smaller the Logarithmic Loss result, the better the prediction.

E. Model training

We use Xgboost and LightGBM to train the training set and make predictions, then use the Logarithmic Loss to evaluate the prediction results. The results obtained are as Figure 6:

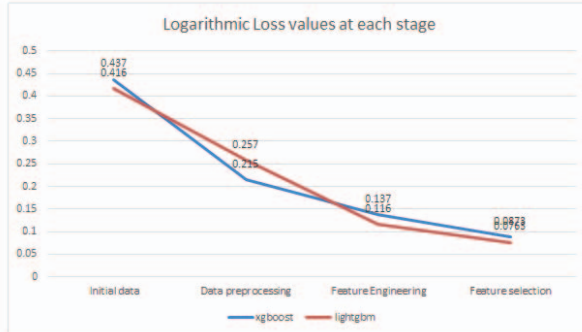

Figure 6. Logarithmic Loss values at each stage

V. CONCLUSION

Mobile advertising has occupied more than half of the online advertising market, and because of its various forms, it also can use the real-time data and historical data to predict the intention of the user, so as to achieve the accurate delivery of advertising. The conversion rate of advertising affects the bidding and final revenue of advertisers, and the accuracy of the prediction of advertising conversion rate has a huge impact on advertisers and advertising platforms. At present, there are many mature results for the conversion rate of search advertising, but the conversion rate of mobile advertising is relatively rare. By using preliminary data Tencent competition provided, through analysis of various factors affecting the AD conversion, and connecting with the real data, we put forward several features, and according to the features establish the model and forecast, and finally got a good result.

In the case of mobile advertising conversion rate, this paper although has achieved some success, but it still has some shortcomings and needs improvement.

(1) There are still deficiencies in the acquisition of features. There are many features of Tencent. This paper uses only a few of them, and the remaining features remain to be discovered and utilized.

(2) In this paper, only used the XGBoost and LightGBM method to predict, but in the field of machine learning, there has kinds of models, how to determine the effectiveness of a model to deal with the problem of mobile AD conversion rate, and get a more accurate prediction results, is still worthy of further research.

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